



More ICT, more creativity? A fsQCA study on configurational pathways to students' creative thinking in PISA 2022

Xiaoyan Chu 

College of Education, Zhejiang University, Hangzhou, China

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ABSTRACT

In the digital era, fostering creative thinking has become a key educational priority, as it enhances learners' ability to solve complex problems and drive innovation. While Information and Communication Technologies (ICTs) are widely adopted in educational settings, their impact on creative thinking remains unclear, largely due to the dominance of linear analytical approaches that fail to capture complex interdependencies. Drawing on Distributed Cognition Theory and the OECD digital competence framework, this study moves beyond traditional models by applying fuzzy-set qualitative comparative analysis (fsQCA) to PISA 2022 data from 81 countries. This configurational approach identifies multiple pathways while capturing the inherent complexity and causal asymmetry of creative thinking in technology-mediated environments. The findings reveal five equifinal configurations, demonstrating that high creative thinking does not emerge from isolated ICT factors but through distinct combinations of ICT, learner, and instructional conditions, exemplified by the synergy of ICT learning resources, learners' self-efficacy, and inquiry-based instructional approaches. Across all high-achieving configurations, ICT self-efficacy consistently emerges as the central condition, while well-integrated instructional design plays a compensatory role when digital access was limited. These results highlight the importance of designing learner-centered environments that foster ICT self-efficacy, encourage collaborative interactions among learners, ICTs, and pedagogy, and promote cohesive integration of ICT into teaching. By shifting the focus from viewing technology as an isolated enabler to emphasizing contextualized, synergistic interventions, this study challenges techno-determinism and proposes a dynamic alignment framework with actionable guidance for customizing ICT integration strategies to foster creative thinking across diverse contexts.

1. Introduction

In the digital era, creative thinking has emerged as a critical skill for navigating complex problems, fostering innovation, and adapting to rapid technological advancements (OECD, 2023; Corrêa & Mourão, 2025). Defined as the ability to analyze problems from multiple perspectives, integrate diverse elements, and generate novel ideas, creative thinking is not a fixed trait but a dynamic cognitive process shaped by environmental interaction, self-determination, and educational experiences (Karwowski et al., 2019; Marcos et al., 2020). Given its situated and malleable nature, creative thinking is particularly responsive to changes in the learning environment. In recent years, with the widespread implementation of digital technologies in education, Information and Communication Technologies (ICTs) have progressively emerged as a key external factor influencing how learners engage with content, tools,

E-mail address: xiaoyanchu@zju.edu.cn.

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and tasks in innovative ways (Jing et al., 2024; Çolpan Güngördü & Yıldırım, 2025). Despite the increasing integration of ICTs in education, the mechanisms through which digital technologies enhance or hinder creative thinking remain unclear (Alzubi et al., 2025; Rabello-Mestre et al., 2025).

The Distributed Cognition Theory (DCT) offers a critical framework for understanding how cognition emerges through the interactions between individuals, artifacts, and environments (Hollan et al., 2000). From this perspective, ICTs are not merely passive tools but active components of the cognitive system that may reshape the process of creative thinking (Alzubi et al., 2025). While the relationship between creativity and technology use is likely reciprocal, DCT provides a strong theoretical rationale for exploring ICT environments as influential structures shaping cognitive outcomes. This view aligns with the PISA 2022 ICT competence framework, which conceptualizes the interplay of ICT accessibility, learner competencies, and instructional integration as enabling conditions for student development (OECD, 2023). The integration of ICT affects not only educational approaches but also the processes of creative engagement (i.e., creative expression, problem-solving, and knowledge creation), which the OECD (2023) defines as central components of creative thinking. However, existing research predominantly relies on linear analytical methods, potentially overlooking the complex, nonlinear, and configurational relationships between ICT and creative thinking (Agasisti et al., 2023; Alzubi et al., 2025).

To address this gap, this study employs fuzzy-set qualitative comparative analysis (fsQCA) to investigate how different configurations of ICT factors are related to students' creative thinking. By analyzing PISA 2022 data through a configurational lens, the analysis extends distributed cognition theory to account for causal asymmetry in technology-enhanced creativity development. Specifically, the following research question is established: **What combinations of ICT-related conditions are associated with high or low levels of creative thinking across countries and regions?**

This study is structured into several key sections. The second section provides an in-depth review of the relevant literature, setting up the conceptual framework that guides the investigation. The third section then outlines the research methodology, detailing the study design, data sources, and analytical procedures. Subsequently, the results and findings are presented and interpreted, with a particular focus on identifying key configurational pathways associated with creative thinking. This is followed by a discussion of the theoretical and practical implications. Finally, the paper concludes with reflections on limitations and directions for future research.

2. Literature review

2.1. Creative thinking

Creative thinking involves analyzing problems from multiple perspectives and integrating diverse elements to generate, evaluate, and refine innovative ideas (Nikolopoulou, 2018; Marcos et al., 2020; OECD, 2023). It is a dynamic cognitive process that translates creative potential into tangible outcomes through interaction with the environment (Karwowski et al., 2019). Three key attributes define creative thinking: environmental interaction, self-determination, and postnatal cultivability. Environmental interaction fosters creative thinking as learners generate novel ideas by exploring their surroundings and leveraging available resources (Nikolopoulou, 2018; Marcos et al., 2020). Moreover, self-determination sets creative thinking apart from general cognition, requiring autonomy in problem construction, idea generation, and self-regulation (Karwowski et al., 2019; Hansen, 2022). A study on middle school students' mathematical problem-solving found that student agency and creative reasoning are closely linked, highlighting the role of autonomy in creativity (Hansen, 2022). In addition, postnatal cultivability emphasizes that creative thinking is not innate but can be developed through structured learning experiences, highlighting the importance of instructional design and learning environments in fostering creativity (Nikolopoulou, 2018).

With the advent of emerging ICTs, the learning environments have undergone substantial changes, reshaping how creative thinking is cultivated in contemporary education (Tang et al., 2022; Mishra et al., 2024). ICTs not only redefine how individuals learn and collaborate, but also open up new avenues for problem-solving and creative expression. Such creative activities are often facilitated by various digital products, including tools for information saving-sharing, virtual simulations, and digital writing (Tang et al., 2022; Alzubi et al., 2025). For instance, a study by Alzubi et al. (2025) involving 546 EFL learners found that generative AI tools are

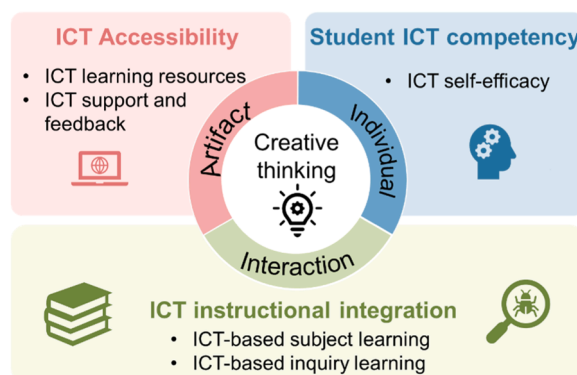


Fig. 1. Mapping ICT and creative thinking: a distributed cognition and PISA 2022 perspective.

conducive to creativity development in second language learning, demonstrating high effectiveness in areas like conversation practice, idea generation, writing feedback, and imaginative thinking. Despite these promising affordances, the mechanisms through which ICTs influence creative thinking remain insufficiently understood.

2.2. The relationship between ICT and creative thinking

As creative thinking is a higher-order cognitive process that interacts with ICTs in complex ways, a multidimensional perspective is needed to fully understand its underlying mechanisms (Tang et al., 2022; OECD, 2023). Distributed Cognition Theory (DCT) challenges the notion of cognition as an isolated psychological process, instead conceptualizing it as the emergent result of interactions among individuals, artifacts, and their environment (Hollan et al., 2000). From this viewpoint, creative cognitive activities are shaped by the intricate interplay among learners, ICT tools and resources, as well as the ICT-supported learning tasks they engage in. Creative thinking is not solely an internal trait but a product of meaningful coordination with external resources. For instance, Zhao (2025), adopting the lens of DCT, effectively sheds light on the innovative role of AI tools as cognitive extensions that enhance students' idea generation and expression during creative writing. This DCT perspective is operationalized by the PISA 2022 ICT assessment framework (see Fig. 1), which models the learning environment as a system of interactions among three core dimensions: ICT accessibility, student ICT competency, and ICT instructional integration (OECD, 2023). Therefore, this framework allows for a theoretically grounded analysis of how specific configurations of ICT conditions are associated with changes in educational approaches and creative engagement processes, which may further shape students' creative thinking.

Firstly, ICT accessibility provides the essential infrastructure and resources (OECD, 2023). Multimedia-rich tools, such as gamified platforms and digital design applications, offer learners diverse opportunities to generate and refine ideas (Tang et al., 2022; Chen et al., 2022). Moreover, immersive virtual environments further enhance creativity by enabling exploration through realistic scenarios and interactive engagement (Chen et al., 2022). Advanced AI-driven ICT systems further amplify this process by delivering instant feedback and adaptive support (Lee & Chung, 2024; Alzubi et al., 2025). For instance, generative AI assists learners in synthesizing concepts into novel solutions, improving problem-solving performance (Urban et al., 2024; Alzubi et al., 2025). An experimental study by Urban et al. (2024) found that AI-enhanced learning environments efficiently increased solution originality and promoted learners' divergent thinking.

Secondly, learners' individual ICT competency significantly impacts how learners harness ICTs for creative thinking. Despite rapid advancements, humans remain central decision-makers in human-computer collaboration (Kim, 2024). Thus, active engagement and strategic use determine the effectiveness of digital tools in fostering creativity. A key factor in this process is ICT self-efficacy, which reflects confidence in using technology (Heo et al., 2021) and has been shown to reduce cognitive load (Redifer et al., 2021), thereby enabling learners to navigate creative tasks more effectively. Additionally, students with greater self-efficacy are also better able to manage technology-related anxiety, maximizing the creative potential of digital tools (Yuan et al., 2024).

Thirdly, the integration of ICT in subject learning and inquiry-based activities plays a crucial role in shaping creative thinking. The PISA 2022 framework identifies ICT instructional integration as a fundamental driver of students' creativity (OECD, 2023). Technology-enhanced multimodal content goes beyond traditional instruction, stimulating creativity by providing dynamic learning experiences (Pérez-Sanagustín et al., 2017; Kumar et al., 2024). Additionally, ICT-supported inquiry learning expands opportunities for independent exploration and creative problem-solving, reinforcing higher-order cognitive skills (Chu et al., 2021).

Although ICTs can serve as cognitive scaffolds that enhance creativity across multiple dimensions, they may also lead to unintended negative outcomes (Alzubi et al., 2025). Over-reliance on ICTs continues to raise concerns, as it may hinder higher-order cognitive skills and potentially stifle creative output (Alzubi et al., 2025). As Oliver (2011) noted, some studies adopt a techno-deterministic view, assuming that ICTs may inherently enhance learning and higher-order thinking. Yet such assumptions often overlook how the actual impact of ICTs is shaped by pedagogical design, learner engagement, and contextual conditions (Díaz & Nussbaum, 2024). This highlights the need for research that considers the complex interplay among technological, instructional, and learner-related factors.

Despite existing insights, research on the relationship between ICTs and creative thinking has largely counted on linear models such as regression analysis (KC et al., 2025) and structural equation modeling (Alzubi et al., 2025). Designed to test fixed causal assumptions, these methods are often inadequate for capturing causal complexity, particularly multiple conjunctural causation (i.e., conditions combining to yield an outcome) and equifinality (i.e., distinct combinations of factors leading to the same result).

2.3. The application of fsQCA in educational research

Fuzzy-set qualitative comparative analysis (fsQCA) is a widely used configurational approach that bridges qualitative contextual depth with the logical rigor of quantitative methods (Ragin, 2009; Pappas & Woodside, 2021). Unlike traditional quantitative techniques that focus on linear relationships, fsQCA employs Boolean algebra and set theory to examine how combinations of multiple conditions lead to an outcome. It identifies commonalities and differences among cases with the same outcomes, constructing concise, multi-factor configurational pathways. Rather than assuming homogeneity across observations, fsQCA is designed to handle case diversity and uncover multiple causal pathways that are contingent on context-specific combinations of conditions (Ragin, 2009; Pappas & Woodside, 2021). This makes fsQCA particularly suitable for research involving heterogeneous samples, multiple interacting variables, and asymmetric causal relationships (Ragin, 2009; Pappas & Woodside, 2021). Based on variable characteristics, QCA includes crisp-set QCA (csQCA) for binary variables, fuzzy-set QCA (fsQCA) for continuous variables, multivalued QCA (mvQCA) for categorical variables, and temporal QCA (tQCA) for sequence analysis. Among these, fsQCA is the most flexible, as it allows for partial

membership degrees, overcoming the rigidity of forced categorization and enhancing analytical precision (Kraus et al., 2018).

FsQCA has been extensively applied to examine instructional design configurations and their effects on learning outcomes (Li et al., 2023; Ma et al., 2025). For example, Ma et al. (2025) demonstrated that different combinations of augmented reality, concept map strategies, and individual differences lead to varying scientific concept learning outcomes, revealing asymmetrical pathways to high and low learning performance. Similarly, when investigating the configurational characteristics of innovative behavior, fsQCA effectively identified multiple distinct pathways (Fréour et al., 2024). Beyond experimental studies, fsQCA has also played a crucial role in analyzing large-scale international assessments, such as TIMSS and PISA, to identify optimal factor configurations influencing student performance (Zheng & Tan, 2024; Chi & Wang, 2024). According to the PISA ICT framework and cognitive distribution theory, creative thinking emerges not as a linear accumulation of factors but as the synergistic interaction of multiple components. The holistic perspective and causal asymmetry analysis of fsQCA make it particularly well-suited for capturing the complexity of such cross-cultural and sociotechnical systems.

3. Methods

3.1. Research design

To investigate the complex configurations of ICT factors associated with learners' creative thinking, this study employs fsQCA for several reasons. First, the development of creative thinking relies on a network of interconnected factors, making fsQCA well-suited to uncover the interrelations and configurational patterns (Misangyi et al., 2017). Second, compared to traditional regression methods, fsQCA can better capture the non-linear and asymmetric relationship between ICTs and creative thinking (Kier & McMullen, 2020). Third, ICT-related factors may form diverse and equivalent pathways linked to different level of creative thinking, which fsQCA reveals more effectively than cluster analysis (Misangyi et al., 2017). Finally, fsQCA provides detailed case-level insights, enabling the verification of individual case specifics and offering more actionable implications for educational practice.

In summary, given the continuous nature of the variables in this study and the need to account for causal complexity, fsQCA was selected as the analytical method. The analysis involves three main steps: (1) necessity analysis of individual conditions, (2) sufficiency analysis of condition configurations, and (3) sensitivity analysis. Data were processed and analyzed using IDB Analyzer and QCA 3.0 software.

3.2. Data sources and samples

PISA is a global assessment that evaluates 15-year-old students' achievement in science, reading, and mathematics, while also gathering extensive data on their personal backgrounds (e.g., their attitudes and beliefs towards science), the instructional experiences (e.g., inquiry-based learning methods), and family contexts (e.g., their socioeconomic status) (OECD, 2023; Chi & Wang, 2024). This study draws from the PISA 2022 data, which covers students' performance and attitudes across 81 participating countries and regions. In response to the global pandemic and the accelerating digital transformation of education, PISA 2022 placed particular emphasis on variables related to digital learning environments, aiming to examine how creative thinking is influenced by ICT-related factors such as access, competence, and integration into educational practice (OECD, 2023). The dataset exhibits substantial heterogeneity across national, school, and individual levels. Countries vary in education systems, ICT infrastructure, and pedagogical models; schools differ in implementation strategies and teacher preparedness; and students report diverse levels of access, self-efficacy, and ICT-based learning experiences (OECD, 2023). To ensure cross-cultural comparability and test equivalence, all assessment tools and materials underwent multiple rounds of review across participating countries (OECD, 2023).

This study examines several key variables that represent ICT integration in digital learning (see Table 1). The outcome variable, creative thinking (PV_CRTH), reflects students' ability to analyze problems, generate novel ideas, and apply innovative solutions. Creative thinking was measured using an Evidence-Centered Design (ECD) approach, which involved both cognitive assessments and student questionnaires. The goal was to understand how students' creative thinking processes manifest during task completion and to explore the relationships with other educational factors. The causal conditions include three main dimensions: ICT accessibility, student ICT competencies, and ICT instructional integration. ICT accessibility is measured by ICT learning resources (ICTRES), which assess the availability of digital tools that support learning, and ICT support and feedback (ICTFEED), which captures the extent of guidance and assistance learners receive when using ICT. Student ICT competencies are represented by ICT self-efficacy (ICTEFFIC), which evaluates learners' confidence in using technology effectively for educational purposes. ICT instructional integration is analyzed

Table 1
Overview of variables and corresponding PISA indicators.

Types	Constructs	Variable names	PISA variable
Outcome variable	Creative thinking	Creative thinking	PV_CRTH
Causal conditions	ICT accessibility	ICT learning resources	ICTRES
		ICT support and feedback	ICTFEED
	Student ICT competencies	ICT self-efficacy	ICTEFFIC
	ICT instructional integration	ICT-based subject learning	ICTSUBJ
		ICT-based inquiry learning	ICTENQ

through ICT-based subject learning (ICTSUBJ), which examines how ICT is incorporated into traditional coursework, and ICT-based inquiry learning (ICTENQ), which explores how ICT supports inquiry-based learning to foster critical thinking and problem-solving skills.

3.3. Data analysis

3.3.1. Data calibration

The fsQCA was conducted at the country level, with each participating country or region treated as a single case in the configurational analysis. To construct the country-level dataset, all student-level variables were first aggregated using the IEA's IDB Analyzer software, following OECD guidelines for PISA data processing (OECD, 2022). This procedure employed plausible values (PVs) and the full set of 80 replicate weights, yielding statistically robust and nationally representative estimates. Given the country-level missing data in the dataset, the multiple imputation method was employed (Rubin, 2018). In the first step, this technique creates multiple complete datasets by imputing missing values based on observed data. These datasets are then analyzed separately and combined to yield more reliable estimates. It is widely recognized as an appropriate approach for managing large-scale survey data, such as PISA (He et al., 2025; Khurma et al., 2025). According to He et al. (2025), missing values in the PISA dataset were imputed using a linear regression model with five iterations.

After imputation, the complete dataset covering 81 countries was calibrated into fuzzy sets using the direct method. This approach allows for the transparent and replicable transformation of raw scores into fuzzy membership values ranging from 0 to 1, where 1 indicates full membership, 0 indicates full non-membership, and intermediate values reflect varying degrees of set membership (Ragin, 2009; Pappas & Woodside, 2021). Calibration anchors were set at 95 %, 50 %, and 5 % of the empirical distribution, which correspond to full membership, the intermediate-set point, and full non-membership, respectively. This percentile-based calibration strategy follows established fsQCA practices (Pappas & Woodside, 2021) and has been widely applied in country-level studies (Nikou et al., 2024; Yagmur et al., 2024). Meanwhile, to ensure the retention of relevant cases, the membership value of 0.5 was adjusted to 0.501 (Cangialosi, 2023). Table 2 presents the descriptive statistics of the original variables prior to calibration. These values are reported to illustrate how the calibration thresholds for fuzzy-set transformation were determined.

3.3.2. Necessity and sufficiency analysis

Following calibration, fsQCA was conducted in two main stages: necessity analysis and sufficiency analysis. The necessity analysis was first performed to examine whether any single condition consistently appeared in cases exhibiting high or low levels of creative thinking. According to established guidelines (Ragin, 2009), conditions with consistency exceeding 0.90 are considered necessary for the outcome and are often treated as "always present" during truth table construction.

After assessing necessity, a truth table was constructed to systematically summarize the relationships between the causal conditions and their corresponding outcomes, forming the basis for subsequent sufficient analysis (Fiss, 2011). The truth table computes all potential configurations, represented by 2^k rows, where k denotes the number of outcome predictors, and each row represents a distinct combination (Pappas & Woodside, 2021).

For truth table rows with adequate empirical evidence, frequency threshold, raw consistency, and proportional reduction in inconsistency (PRI) were applied to identify significant configurations (Ragin, 2009). Firstly, the frequency threshold refers to the minimum number of cases required for a condition combination to be included in the table. It serves to exclude rare configurations that are based on too few cases to yield meaningful insights (Pappas & Woodside, 2021). In line with recent fsQCA studies using moderate-sized samples (fewer than 150 cases) (Soliman et al., 2023; Xu & Xu, 2025), we set the frequency threshold at 2. Secondly, raw consistency indicates how reliably a specific combination of conditions leads to the desired result, based on the proportion of cases that exhibit both the configuration and the outcome (Pappas & Bley, 2023). Following commonly accepted practices (Pappas et al., 2019; Hu et al., 2025), we adopted a threshold of 0.80 to determine sufficient consistency. Thirdly, the PRI value ensures that configurations are not simultaneously subsets of both the outcome and its absence, thereby enhancing the directionality and causal clarity of the findings (Pappas & Bley, 2023). PRI threshold at 0.65 was set according to widely recognized protocols (Hu et al., 2025; Xu & Xu, 2025). Configurations that satisfy all three thresholds are deemed sufficient for the desired outcome, with the outcome column in the truth table marked as 1 in these cases, and 0 otherwise. A truncated version of the truth table for high creative thinking is provided in

Table 2

Variable calibration anchors and descriptive statistics of original variables.

Variable name	Calibration anchors			Descriptive statistics		
	Full membership	Intermediate-set point	Full non-membership	SD	Min	Max
Creative thinking	37.25	28.83	14.59	6.77	40.93	11.18
ICT learning resources	0.62	-0.63	-1.63	0.74	0.81	-2.47
ICT support and feedback	0.44	0.05	-0.42	0.25	0.73	-0.53
ICT self-efficacy	0.23	-0.08	-0.41	0.19	0.28	-0.54
ICT-based subject learning	0.73	-0.04	-0.51	0.36	1.32	-1.24
ICT-based inquiry learning	0.49	-0.02	-0.53	0.29	0.8	-0.82

Note. Calibration anchors were set at the 95th percentile (full membership), 50th percentile (crossover point), and 5th percentile (full non-membership) of the original variable distributions. Descriptive statistics refer to the uncalibrated values.

Appendix A for reference.

Subsequently, Boolean algebra was employed in the configuration analysis to identify consistent patterns of conditions that were associated with the desired outcome (Kumar et al., 2022), thereby uncovering multiple pathways to the development of creative thinking. Logical minimization was applied to the truth table to derive the solutions, which consisted of either individual conditions or combinations of conditions, that could predict students' creative thinking.

3.3.3. Sensitivity analysis

To assess the robustness of the identified configurations, we conducted a sensitivity analysis focusing on both calibration anchors and threshold values (Fiss, 2011; Mattke et al., 2022). For calibration, we replicated the analysis using alternative anchor points (80 %, 50 %, 20 %) and followed the recommended procedure of subtracting a small constant from the crossover value (0.50), rather than only adding one. For threshold sensitivity, we adjusted the frequency threshold from 2 to 1. In addition, the raw consistency threshold was increased from 0.80 to 0.85, and the PRI threshold from 0.65 to 0.75 to examine whether the configurational solutions remained stable across different parameter settings.

4. Results and discussion

4.1. Necessity analysis of individual condition

In this study, necessity analysis was performed using consistency and coverage as core indicators to assess whether individual variables constitute necessary conditions for explaining the outcomes. Consistency reflects the proportion of outcome cases in which a condition variable appears, while coverage measures the strength of the explanatory power of a condition variable (Pappas & Woodside, 2021). As shown in Table 3, the consistency of individual condition variables did not exceed the reference value of 0.9, suggesting that no single variable alone can sufficiently explain the role of ICT in shaping learners' creativity.

4.2. Sufficient analysis of condition configuration

When no single variable qualifies as a necessary condition, sufficient analysis further examines different combinations of condition variables (configurations) to assess their sufficiency in producing a specific outcome. Based on the consistency, PRI, and frequency thresholds defined in the method section, three types of configurations that sufficiently explained the outcome were identified: parsimonious, intermediate, and complex. Among them, the intermediate solution was prioritized for its balance between empirical richness and theoretical relevance (Fiss, 2011; Pappas & Woodside, 2021). Unlike the complex solution, which includes all possible configurations and can be challenging to interpret, or the parsimonious solution, which risks oversimplification by retaining only core conditions, the intermediate solution incorporates conditions strongly supported by prior theory and empirical knowledge. It presents both core and peripheral conditions, making the results more meaningful and easier to interpret (Fiss, 2011; Pappas & Woodside, 2021). The parsimonious solution was also referenced to identify essential core conditions consistent across configurations. In this context, consistency and coverage also serve as key indicators of a configuration's explanatory power. Consistency reflects how reliably the outcome occurs when a configuration is present, while coverage, comparable to R-squared in regression, indicates the extent to which the configuration accounts for the outcome (Pappas & Woodside, 2021; Mattke et al., 2022).

As shown in Table 4, five configuration pathways were identified for achieving high creative thinking, with an overall consistency of 0.896 and an overall coverage of 0.682, accounting for 68.2 % of cases with high creative thinking at the national level. Meanwhile, two configuration pathways were identified for low creative thinking, with an overall consistency of 0.886 and an overall coverage of 0.481, explaining 48.1 % of cases with low creative thinking at the national level. These results suggest a robust explanatory power.

4.2.1. Configurations of high creative thinking

As shown in Table 4, five pathways (C_h1 to C_h5) lead to high creative thinking:

Table 3

Results of necessity analysis for individual condition variables.

Condition variable	High creative thinking		Low creative thinking	
	Consistency	Coverage	Consistency	Coverage
ICT learning resources	0.627	0.693	0.594	0.576
~ICT learning resources	0.617	0.633	0.684	0.617
ICT support and feedback	0.674	0.697	0.670	0.608
~ICT support and feedback	0.621	0.682	0.666	0.642
ICT self-efficacy	0.790	0.864	0.507	0.487
~ ICT self-efficacy	0.531	0.551	0.858	0.782
ICT-based subject learning	0.641	0.748	0.600	0.615
~ ICT-based subject learning	0.670	0.656	0.754	0.648
ICT-based inquiry learning	0.660	0.688	0.687	0.629
~ ICT-based inquiry learning	0.644	0.701	0.659	0.630

Note: The symbol ~ represents logical negation.

Table 4

Results of sufficient analysis for condition configuration.

Condition Variable	High creative thinking configurations					Low creative thinking configurations	
	C _h 1	C _h 2	C _h 3	C _h 4	C _h 5	C _l 1	C _l 2
ICT learning resources	●	●	⊗			⊗	●
ICT support and feedback	⊗				●	●	●
ICT self-efficacy	●	●	●	●	●	⊗	⊗
ICT-based subject learning	⊗		●	⊗	●	⊗	●
ICT-based inquiry learning		●	●	●	⊗	●	●
Consistency	0.975	0.923	0.914	0.939	0.936	0.867	0.911
Raw coverage	0.305	0.458	0.384	0.439	0.371	0.36	0.35
Unique coverage	0.020	0.079	0.045	0.036	0.005	0.13	0.12
Example countries/ regions	Costa Rica, Spain, Slovenia	Denmark, Norway, United Kingdom, Macao (China), United States, Canada, Switzerland, etc.	New Zealand, Italy, Hong Kong (China), Ireland, Peru, etc.	South Korea, United Kingdom, Hong Kong (China), Spain, Belgium, Lithuania, etc.	Singapore, New Zealand, Ireland, France, Malta, Turkey, etc.	Indonesia, Philippines, Panama, Georgia, Vietnam, etc.	Bulgaria, Thailand, Uzbekistan, Romania, Mexico
Solution consistency	0.896					0.886	
Solution coverage	0.682					0.481	

Note: ● indicates the presence of a core condition; ⊗ indicates the absence of a core condition; ● indicates the presence of a peripheral condition; ⊗ indicates the absence of a peripheral condition; blank cells indicate that the presence or absence of the condition has no impact on the outcome.

C_h1: This configuration is characterized by learners' ICT self-efficacy as the core condition, supplemented by ICT learning resources that support creative thinking. This pathway suggests that even when ICTs provide limited support and feedback in course learning, and regardless of their ability to facilitate inquiry-based learning, learners with high ICT self-efficacy can still maximize the use of ICT resources to assist their creative process. Among the participating countries in PISA 2022, three countries fit this configuration. Taking Spain as an example, learners generally exhibit a strong sense of self-efficacy, which plays a crucial role in fostering creative learning activities (Zammiti et al., 2023). Supporting this, Spain ranks among the top EU countries in digital connectivity and public digital services. Under the framework of the *Digital Education* initiative, the country has invested €260 million to equip educational institutions with digital tools and enhance digital connectivity, significantly improving access to digital learning resources (Ministry of Education & Vocational Training, 2020). This robust digital infrastructure further reinforces the creative potential of students with high self-efficacy.

C_h2: This configuration features ICT self-efficacy as the core condition, combined with ICT learning resources and ICT-facilitated inquiry-based learning. This pathway comprehensively covers three dimensions: access to ICT, students' ICT competencies, and ICT instructional integration, aligning with the OECD digital competence framework and the distributed nature of cognition. Also, the lack of any absent condition suggests that this configuration consists entirely of positively contributing elements, enhancing its feasibility. Among the configurations related to high creative thinking, this pathway demonstrates the highest coverage, encompassing 15 countries and regions, including the United States, the United Kingdom, and Macao (China), etc. These features collectively make C_h2 a compelling optimal state in terms of both theoretical coherence and practical generalizability. The balanced and scalable integration of ICT access, learner competence, and inquiry-oriented pedagogy helps explain its global prevalence, as these elements are increasingly emphasized in recent educational reforms (Wang et al., 2024) and can be implemented without requiring extensive curriculum redesign or advanced ICT support (Abedi, 2024). Taking the US as an example, since the launch of the *National Educational Technology Plan* in the late 20th century, the country has prioritized student competence development and advocated inquiry-driven computer-supported learning resources. When integrating ICTs pedagogically, learners in the US are often encouraged to engage in interdisciplinary inquiry through STEM or project-based approaches, which jointly serve to cultivate their creative thinking (Martinez, 2022).

C_h3: This configuration centers on ICT self-efficacy while emphasizing the seamless integration of ICT in both subject learning and inquiry-based learning. This pathway highlights the critical role of instructional design and educational applications in leveraging ICT for creative thinking development. Even in contexts where ICT resources are relatively scarce or provide limited support, the effective combination of ICT with course learning and inquiry-based learning can significantly benefit high creative thinking learners. Countries and regions fitting this configuration in PISA 2022 include New Zealand, Italy, and Hong Kong (China), etc. For instance, despite intense academic competition and limited human and facility resources in schools compared to the OECD average, Hong Kong has maintained high educational standards through a strong school quality assurance system. By investing significantly in curriculum development and assessment, Hong Kong has emphasized the integration of structured instruction with inquiry-based learning, creating an environment that nurtures students' creative thinking (Weng & Chiu, 2024; Su & Yang, 2024).

C_h4: This configuration is built upon ICT self-efficacy as the core condition, complemented by ICT-facilitated inquiry-based learning, suggesting a mutually reinforcing relationship conducive to creative engagement. This pathway underscores the crucial role of ICT-supported inquiry learning in fostering creative thinking among learners with high ICT self-efficacy. Countries fitting this configuration in PISA 2022 include South Korea, the United Kingdom, Hong Kong (China), and Spain, etc. For instance, the UK Department for Education (2019) prioritizes digital competency development through its *Realising the Potential of Technology in Education* framework, emphasizing user-centered service design. Complementing this, Arts Council England's *Creativity Collaboratives* (2021) focuses on pedagogical innovation through interdisciplinary inquiry-based approaches that nurture creative cognition and divergent thinking capacities.

C_h5: This configuration features ICT self-efficacy as the core condition, with ICT support and feedback as well as ICT-facilitated subject learning serving as peripheral conditions in a configuration sufficient for high creative thinking. In addition to ICT self-efficacy, which serves as a core condition across all pathways, C_h5 emphasizes the importance of effective ICT support and feedback, as well as the application of ICT in course learning. Countries fitting this configuration include Singapore, New Zealand, and Ireland, etc. Notably, Singapore's *Framework for 21st Century Competencies and Student Outcomes* underlines the importance of integrating ICT skills development within academic curricula (Ministry of Education, Singapore, 2024). Professional communities comprising curriculum advisors, teachers, and academic administrators collaborate to explore future-oriented curriculum content, teaching models, and technology-supported solutions, providing strong support for the development of 21st-century skills, including creative thinking (Glatthorn et al., 2018). Additionally, analysis of PISA mathematics and reading data also reveals that New Zealand students in this configuration exhibited creative thinking levels surpassing predicted benchmarks. This achievement stems from the strategic alignment of digital transformation initiatives with curricular development, combined with the integration of digital competencies into tertiary admission criteria, which collectively establish a robust framework for nurturing innovators with dual expertise in technical proficiency and creative problem-solving (Thelma et al., 2024).

4.2.2. Configurations of low creative thinking

Two pathways leading to low creative thinking are identified as C_l1 and C_l2:

C_l1: This configuration is characterized by the absence of three core conditions, including ICT resources, ICT self-efficacy, and ICT-facilitated subject learning. Without these foundational elements, fostering students' creative thinking remains challenging, even when inquiry-based learning approaches are adopted. Countries fitting this configuration include Indonesia, the Philippines, Panama, Vietnam, etc. For instance, Indonesia has introduced national initiatives such as the "*Merdeka Belajar*" (Freedom to Learn) curriculum

to promote active and student-centered digital learning. However, limited ICT access has undermined students' development of ICT literacy and self-efficacy, leaving them unprepared to engage in creative and exploratory tasks (Lestari et al., 2024). In PISA 2022, Indonesia scored a mean of 19.0 in creative thinking, ranking between 56th and 59th among all participating countries and economies, reflecting relatively low performance (OECD, 2024). In this context, the lack of digital readiness is often compounded by infrastructural challenges, leaving students without the essential conditions needed to foster high creative engagement.

C₂: This configuration illustrates that although supportive digital learning environments may be available, including ICT-facilitated subject learning and effective ICT feedback mechanisms, the absence of ICT self-efficacy still constrains students' ability to benefit from these resources. Countries aligned with this configuration include Bulgaria, Thailand, Uzbekistan, etc. Taking Thailand as an example, national policies have actively promoted classroom digitalization and ICT infrastructure development (United Nations Thailand, 2021). However, Thailand ranked around 56th in creative thinking in PISA 2022, reflecting relatively low performance compared to other participating countries (PISA, 2024). A recent regression analysis in Thailand also found that students lacking digital confidence are less likely to take advantage of ICTs effectively for academic purposes, limiting opportunities for creative engagement (KC et al., 2025).

4.3. Sensitivity analysis

Sensitivity analysis results suggested that the configurational pathways for both high and low creative thinking were largely stable across all variations in calibration and threshold settings. While minor changes were observed in peripheral conditions, all core causal conditions persisted across the tested models. Solution consistency ranged from 0.840 to 0.900 for high creative thinking and from 0.865 to 0.911 for low creative thinking. Coverage ranged from 0.671 to 0.739 for high creative thinking and from 0.350 to 0.602 for low creative thinking. These values are within the commonly accepted thresholds for sufficiency and empirical relevance in fsQCA (Pappas & Woodside, 2021). Notably, the consistency and coverage values in the solution identified in this study (0.896 and 0.682 for high creative thinking; 0.886 and 0.481 for low creative thinking) fall toward the upper end of the observed robustness ranges. This further reinforces the credibility and robustness of the identified configurations.

4.4. Theoretical implications

Firstly, this study offers a critical perspective on techno-deterministic assumptions by showing that increased access to ICTs alone does not necessarily lead to improved creative thinking outcomes (Oliver, 2011). Across the identified configurations, no single ICT condition emerged as necessary, and several pathways to low creative thinking were characterized by the absence of ICT self-efficacy, despite the presence of other digital resources. These findings challenge the notion that technology inherently drives cognitive gains and instead highlight the central role of learner agency (Hansen, 2022; Malik et al., 2025). In doing so, the study supports a shift toward more interactionist and socio-technical frameworks that account for how learners actively engage with and make meaning through technology, rather than passively benefiting from its availability (Yang et al., 2024).

Secondly, the central role of ICT self-efficacy in our findings offers important theoretical insights in relation to both Distributed Cognition Theory and the PISA 2022 ICT competence framework. This aligns with broader research highlighting self-efficacy as a key antecedent for creative outcomes (Yuan et al., 2024), which our study confirms and further elaborates within ICT and digital learning contexts. Echoing the core assumption of Distributed Cognition Theory, ICT self-efficacy reflects learners' ability to coordinate meaningfully with tools and environments, highlighting the triadic interaction between individuals, artifacts, and contexts (Hollan et al., 2000). Similarly, although the PISA ICT framework includes access, instructional integration, and self-efficacy (PISA, 2023), our findings (e.g., C₁, C₂) extend this framework by demonstrating that, without strong self-efficacy, access and integration alone are insufficient to foster creative outcomes.

Thirdly, the use of fsQCA in this study contributes to theory-building by revealing how different combinations of ICT-related conditions collectively shape creative thinking. Rather than treating ICT as a uniform enhancer (Tang et al., 2022), the analysis shows that creativity emerges through the structural alignment of access to digital tools, learner competence, and pedagogical practices. The most prevalent high-performing pathway (C_{h2}) exemplifies this integrative model by showing how creativity arises from the coordination of resources, self-efficacy, and inquiry-based learning. This finding extends existing frameworks by emphasizing the need for more integrated models of technology-mediated creativity, highlighting the importance of the interaction between technological affordances and pedagogical strategies (Díaz & Nussbaum, 2024).

4.5. Practical implications

4.5.1. Strengthening learner-centered instructional design

Given the pivotal role of ICT self-efficacy in shaping creative thinking pathways, learner-centered instructional design and autonomous, innovation-driven learning are essential. ICT self-efficacy can be strengthened through three key strategies: teacher feedback, vicarious experiences, and self-regulation (Liu et al., 2021b; Kundu, 2020; Srisupawong et al., 2018). First, within ICT-integrated teaching activities, educators should cultivate positive learning experiences. By guiding students to achieve success in using ICT for creative tasks and providing constructive feedback, teachers can effectively enhance learners' confidence in their technological capabilities (Liu et al., 2021b). Second, modeling best practices in ICT-supported creative thinking can reinforce self-efficacy. Observing role models engage in digital problem-solving helps students anticipate potential challenges, develop adaptive strategies, and build resilience in navigating complex learning environments (Kundu, 2020). Finally, fostering learner autonomy in

digital environments further strengthens ICT self-efficacy by promoting self-regulation and independent management of creative endeavors (Srisupawong et al., 2018). Integrating these strategies into pedagogical design not only enhances students' confidence in using technology but also nurtures their capacity for sustained creative engagement.

4.5.2. Integrating multi-dimensional factors

National digital education policies often focus on hardware investment (e.g., the EU's *Digital Education Action Plan*). However, the configurational findings of this study highlight the need for a policy mix approach, emphasizing a more integrated and strategic design. Depending on the specific educational context, various pathways can be employed to integrate different ICT dimensions in fostering students' creative thinking. Following C_h2, as seen in the United States, the United Kingdom, and Macau, students should be provided with ample ICT resources and encouraged to engage in inquiry-based learning. With the growing integration of large language models in education, AI-Generated Content (AIGC) offers powerful tools for producing creative learning materials (Huang et al., 2024). These resources help learners visualize ideas, explore alternative representations, and collaboratively construct knowledge, supporting core processes involved in creative thinking. Moreover, according to C_h5, as seen in Singapore and New Zealand, strengthening ICT-supported feedback mechanisms can mitigate resource constraints while tightly integrating technology with creativity-focused curricula. For instance, Singapore's model of integrating technology development with curriculum design incorporates information skills and creative thinking into national educational objectives (Ministry of Education, Singapore, 2024).

4.5.3. Enhancing precise pedagogical integration

When ICT resources are limited, precise pedagogical design can facilitate the development of student's creative thinking by aligning closely with curriculum content and supporting inquiry-based learning. In curriculum design, ICT tools can be used to promote idea generation, which is an essential component of creative thinking, by enabling students to engage in tasks such as imagining, predicting, and designing within their subject areas, thereby fostering creativity (Lee, 2018). In inquiry-based learning activities, ICT tools support various learning formats, including authentic, guided, collaborative, collective, and game-based inquiry. These formats provide students with greater opportunities for exploration and innovation (Liu et al., 2021a). Furthermore, by providing scaffolding mechanisms to assist students in learning tasks, ICT tools promote active participation and deep engagement, thereby enhancing comprehension and creative problem-solving abilities (Tang et al., 2022).

5. Conclusion

The cultivation of creative thinking is not only a key 21st-century skill but also essential for talent development in a society increasingly shaped by emerging ICTs. This study employs fsQCA to explore the complex configurations of factors influencing creative thinking, focusing on three dimensions: ICT accessibility, students' ICT competence, and ICT instructional integration. The findings highlight that students' ICT self-efficacy plays a central role in fostering creative thinking. While providing comprehensive ICT resources and applications is beneficial, well-designed pedagogical strategies can also achieve the goal of enhancing creative thinking, even in resource-limited environments.

The findings from this study provide valuable insights, but there are several limitations that should be addressed in future research. Firstly, although supported by a theoretical framework, the use of cross-sectional data limits the ability to infer temporal dynamics. The possibility of reciprocal or reverse influences between ICT use and creative thinking cannot be ruled out. Future studies could adopt longitudinal or experimental designs to more robustly examine the directionality of this relationship. Secondly, this study is constrained by its reliance on a single secondary data source. While PISA offers rich cross-national data, the available variables may not be comprehensive and could have overlooked other important aspects. Future research could incorporate additional theoretical perspectives, collect primary data (e.g., surveys or interviews), and employ mixed-method approaches to gain a more comprehensive understanding. Thirdly, this study focuses on secondary school students, leaving open the question of whether the mechanisms and effects identified apply to learners at different educational stages. Developmental differences in cognitive and digital readiness across age groups may moderate the observed patterns. Therefore, future studies should examine whether the identified configurations hold in primary or tertiary education contexts. Finally, although efforts were made to improve measurement and analysis, the influence of data heterogeneity remains. Variations in national policies, school support, and students' ICT readiness may affect the findings across cultural contexts. Future research could incorporate culture-related variables or conduct cross-cultural comparisons to better account for these differences.

CRediT authorship contribution statement

Xiaoyan Chu: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Appendix A

Table A1 presents a truncated version of the truth table that illustrates the main configurations for high creative thinking.

Table A1

Truncated truth table for high creative thinking.

ICTRES	ICTFEED	ICTEFFIC	ICTSUBJ	ICTENQ	Outcome	Cases	Raw consistency	PRI
1	0	1	1	1	1	Finland, Portugal, Switzerland, United States	0.99	0.96
1	0	1	0	1	1	Costa Rica, Spain	0.98	0.92
1	0	1	0	0	1	Chinese Taipei, Slovenia	0.98	0.92
1	1	1	1	0	1	France, Singapore, Ukrainian regions (18 of 27)	0.97	0.90
1	1	1	0	1	1	Croatia, Macao (China), United Kingdom	0.95	0.85
0	0	1	0	1	1	Baku (Azerbaijan), Belgium	0.95	0.83
0	0	1	1	0	1	Estonia, Hong Kong (China), Italy, Kazakhstan, Latvia, Peru	0.94	0.85
0	1	1	1	0	1	Ireland, Kosovo, Malta, New Zealand, Turkey	0.93	0.82
0	1	1	0	1	1	Korea, Lithuania	0.93	0.80
1	1	1	1	1	1	Canada, Chile, Denmark, Norway, Serbia, Sweden	0.92	0.80

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